

REBOLY, Russian Federation

WMO: 226020

Lat: 63.833N

Lon: 30.817E

Elev: 597

StdP: 14.38

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-16.5	-27.6	1.2	-22.0	-21.6	1.7	-16.4	14.4	25.3	13.9	25.1	0.8	250	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	78.4	63.8	74.9	62.1	71.7	60.3	66.2	73.8	64.0	72.1	62.0	69.4	5.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.3	89.1	69.5	60.9	81.7	67.4	58.7	75.5	65.5	31.2	73.6	29.5	72.3	28.1	69.4	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	12.2	11.2	-26.9	83.1	6.6	4.6	-31.6	86.4	-35.5	89.1	-39.2	91.6	-44.1	95.0		
(5)			-27.1	68.5	6.1	4.2	-31.5	71.5	-35.0	73.9	-38.5	76.3	-42.9	79.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	12.7	13.3	21.9	33.4	45.4	55.3	61.3	57.1	48.3	37.1	26.5	19.1		
	DBStd	19.25	13.97	14.68	10.03	8.17	8.76	7.15	6.21	5.91	6.15	7.19	9.25	12.92		
	HDD50	5953	1156	1027	870	499	195	26	2	8	104	403	704	959		
	HDD65	10609	1621	1447	1335	948	608	302	149	253	502	866	1154	1424		
	CDD50	876	0	0	0	1	53	185	353	230	52	2	0	0		
	CDD65	57	0	0	0	0	2	12	34	9	0	0	0	0		
	CDH74	376	0	0	0	0	20	81	216	59	0	0	0	0		
	CDH80	46	0	0	0	0	1	10	29	6	0	0	0	0		
	WSAvg	5.0	5.0	5.0	5.2	4.7	5.1	5.0	4.8	4.4	5.1	5.3	5.4	5.2		
	PrecAvg	23.5	1.3	1.2	1.2	1.4	1.8	2.3	2.8	3.0	2.6	2.3	1.9	1.6		
(15) Precipitation	PrecMax	31.7	2.6	2.3	2.9	3.9	4.7	5.6	5.1	6.1	5.7	5.1	4.8	3.1		
	PrecMin	15.6	0.3	0.1	0.1	0.2	0.0	0.8	0.6	0.5	0.6	0.6	0.6	0.4		
	PrecStd	3.7	0.6	0.6	0.7	0.8	1.0	1.1	1.1	1.3	1.1	1.1	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.5	37.4	46.3	61.5	77.0	81.1	84.0	80.2	68.4	55.0	45.3	38.2		
		MCWB	35.0	34.5	38.4	48.3	58.6	63.7	66.6	64.7	57.9	51.6	43.7	37.3		
	2%	DB	33.8	34.7	40.4	53.4	70.5	75.7	79.7	75.0	63.7	51.2	42.5	35.8		
		MCWB	32.4	33.1	34.3	43.7	55.4	61.2	65.0	62.8	55.9	48.9	41.0	34.5		
	5%	DB	32.5	33.0	36.8	48.9	65.0	71.7	76.2	71.5	60.3	49.0	39.7	34.1		
		MCWB	31.4	31.9	32.7	40.4	52.2	59.3	63.6	61.0	54.4	47.0	38.5	32.9		
	10%	DB	30.9	31.6	34.4	45.1	60.3	68.1	72.8	67.8	57.6	46.7	37.1	32.8		
		MCWB	29.9	30.8	31.4	38.4	49.7	57.4	61.8	59.5	53.2	44.7	36.0	31.8		
	0.4%	WB	35.4	35.2	38.7	49.5	61.2	67.3	70.2	67.5	60.1	52.0	44.7	37.6		
		MCDB	36.2	36.3	45.5	58.4	72.7	73.8	79.2	75.7	65.3	53.7	45.1	37.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.0	33.3	35.3	45.1	57.1	63.6	67.2	64.4	57.6	49.5	41.3	35.0		
		MCDB	33.6	34.3	39.1	52.2	66.8	72.8	75.3	71.9	61.5	50.8	42.3	35.6		
	5%	WB	31.7	32.2	33.5	41.5	53.9	61.1	65.2	62.2	55.7	47.4	38.6	33.2		
		MCDB	32.4	32.9	35.9	47.2	62.7	69.8	73.3	69.0	58.8	48.8	39.4	33.8		
	10%	WB	29.9	30.5	32.0	39.0	50.7	58.3	63.3	60.4	53.9	44.9	36.1	31.9		
		MCDB	30.9	31.5	34.2	44.2	58.7	65.6	70.8	66.7	57.0	46.2	37.0	32.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	13.0	15.0	15.5	15.2	14.3	14.0	13.3	10.7	7.0	7.4	9.5		
		MCDBR	6.7	9.1	15.0	21.1	24.4	19.6	19.1	19.3	15.2	8.7	7.0	7.0		
		MCWBR	6.8	8.8	11.1	12.7	12.4	8.8	8.1	8.6	9.3	7.0	6.9	6.7		
		MCDBR	6.3	8.6	11.8	18.8	21.0	17.9	16.6	16.4	11.8	8.4	6.7	6.6		
	5% WB	MCWBR	6.6	8.4	9.5	12.1	11.9	9.1	8.2	8.9	8.4	7.5	6.8	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.217	0.279	0.307	0.342	0.348	0.344	0.371	0.349	0.326	0.304	N/A	0.205		
	taud		1.961	2.095	2.142	2.254	2.445	2.485	2.427	2.467	2.516	2.385	N/A	1.953		
	Ebn at Noon		113	198	245	264	274	278	265	261	245	196	N/A	49		
	Edh at Noon		9	21	30	34	31	30	31	27	21	16	N/A	4		
(44) All-Sky Solar Radiation	RadAvg		52	230	693	1178	1484	1638	1541	1157	643	233	54	20		
	RadStd		6	48	76	112	179	111	163	117	69	35	10	2		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (20 neighbors)	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	-346	N/A	N/A	

Nomenclature: See separate page